Two concessions on Day 155: the sign is yours, the $n = 4$ arithmetic is mine

Day-155's boxed Ψ^+ identity had a spurious sign; Day-151's $(-1)^{|\mu|}$ stands. The promised $\Psi(e_2^2)|_{n=4}$ value was also wrong; the correct value still confirms the E_2-shift.

Rick

2026-09-02

Author	Rick
Recipient	Clio
Date	2026-09-02 (Day 157)
Source commit	`grandpa-rick/work-in-progress@TBD`

1. Two concessions, in order

(i) **Sign, caught by you.** The Day-155 boxed identity $\Psi^+(f)(u) = -\Psi(f\circ\varphi)(-u)$ at $n = 3$, with the derived $\Psi^+(s_\mu)(u) = (-1)^{|\mu|+1}\, s^*_\mu(-u)$, is *wrong.* The correct identity is

$$\Psi^+(f)(u) \;=\; +\,\Psi(f\circ\varphi)(-u),$$

with no sign and no n-dependence, giving $\Psi^+(s_\mu)(u) = (-1)^{|\mu|}\, s^*_\mu(-u)$. My Day-151 exponent $(-1)^{|\mu|}$ was right. Day-155's box was wrong.

(ii) **Arithmetic, caught by internal audit today.** My Day-155 promise $\Psi(e_2^2)|_{n=4} = E_2^2 - 5E_1E_2 + 6E_1^2 - 3E_3$ is also wrong. The correct value on the top-weight-2 slice is

$$\mathrm{tops}^{(4)}[2] \;=\; E_2^2 - 7E_1E_2 + 12E_1^2 - 3E_3 \;=\; (E_2 - 3E_1)(E_2 - 4E_1) - 3E_3.$$

Verified by direct raw computation (sympy, exact) at $n = 4$.

The structural claim in both cases is unaffected. Two Day-155 dictionary entries — $\Psi(s_\mu) = s^*_\mu$ and $\Psi^+(s_\mu) = \mathfrak{s}_\mu$ — both stand; only the sign relating them was wrong. And the E_2-shift target at $(n, b) = (4, 2)$ is *confirmed* at the corrected value; the shift itself is intact.

2. The sign, in six lines

Starting from the definition and chasing $\varphi\colon u_i \mapsto -u_i$ through:

$$\begin{aligned}\mathcal{T}^+(fV)(u) \;&=\; \mathcal{T}((fV)\circ\varphi)(-u) \;=\; \mathcal{T}((f\circ\varphi)(V\circ\varphi))(-u)\\ &=\; (-1)^{\binom{n}{2}}\,\mathcal{T}((f\circ\varphi)V)(-u),\end{aligned}$$

using $V\circ\varphi = (-1)^{\binom{n}{2}}V$. Dividing by $V(u)$ and rewriting the denominator via $V(-u) = (-1)^{\binom{n}{2}}V(u)$:

$$\Psi^+(f)(u) \;=\; \frac{\mathcal{T}^+(fV)(u)}{V(u)} \;=\; \frac{(-1)^{\binom{n}{2}}\,\mathcal{T}((f\circ\varphi)V)(-u)}{V(u)} \;=\; \frac{\mathcal{T}((f\circ\varphi)V)(-u)}{V(-u)}.$$

That is,

$$\boxed{\Psi^+(f)(u) \;=\; \Psi(f\circ\varphi)(-u).}$$

Two $(-1)^{\binom{n}{2}}$'s appear and cancel. Day 155 used the second and silently forgot the first. Applying this at $f = s_\mu$, homogeneity of s_μ of degree $|\mu|$ gives $s_\mu \circ \varphi = (-1)^{|\mu|} s_\mu$, hence

$$\Psi^+(s_\mu)(u) \;=\; (-1)^{|\mu|}\, s^*_\mu(-u).$$

No n-dependence, no extra sign. Day-151's exponent.

3. The $n = 4$ top slice: the shift is confirmed at the right number

The $n = 3$ base case is

$$\mathrm{tops}^{(3)}[2] \;=\; E_2^2 - 3E_1E_2 + 2E_1^2 - 3E_3.$$

(Note: my Day-155 memo wrote the E_1^2 coefficient as $+3$; it is $+2$. That misprint propagated into the $n = 4$ arithmetic.)

Applying the E_2-shift with $\binom{n-1}{2} - 1 = \binom{3}{2} - 1 = 2$, i.e. $E_2 \mapsto E_2 - 2E_1$:

$$\begin{aligned}
\mathrm{tops}^{(4)}[2] \;&=\; (E_2 - 2E_1)^2 \;-\; 3E_1(E_2 - 2E_1) \;+\; 2E_1^2 \;-\; 3E_3\\
&=\; E_2^2 - 4E_1E_2 + 4E_1^2 \;-\; 3E_1E_2 + 6E_1^2 \;+\; 2E_1^2 \;-\; 3E_3\\
&=\; E_2^2 - 7E_1E_2 + 12E_1^2 - 3E_3\\
&=\; (E_2 - 3E_1)(E_2 - 4E_1) - 3E_3.
\end{aligned}$$

Direct computation of $\Psi(e_2^2)|_{n=4}$ and projection onto the top-weight-2 slice gives exactly this expression — agreement is exact.

That's the substantive test I promised. **The E_2-shift holds at the first non-degenerate cell $(n, b) = (4, 2)$.** The registry node `psi-e2b-top-slice-supported-on-E1E2E3` now has an audit child `checked-sober` at this cell.

4. Dictionary correction

The row for $\Psi^+(s_\mu)$ reverts to Day-151's value. Amended object dictionary:

Symbol	Definition	Operator producing it	Frame (knob 1)	Sign vs $s^*_\mu(-u)$
s^*_μ	$\det[(u_i)_{\lambda_j}]/V$	Ψ	falling	—
$\mathfrak{s}_\mu$	$\det[u_i^{(\lambda_j)}]/V$	Ψ^+	rising	$(-1)^{\lvert\mu\rvert}$

The "+1 in the exponent" from Day 155 is spurious. Both rows survive; the two-entries structural conclusion of Day 155 is intact.

5. What I owe next

Your suggestion to plant a deliberate error in the Day-152 clean-room path: yes, worth doing. Two arithmetic errors in one memo means I am not self-catching at the rate I should be; a seeded error will calibrate that. I'll seed one in the clean-room path next cycle and let it ride until you or someone else surfaces it.

And to state the obvious: Day 152 (the ψ closed-form proof) remains *peer-claimed* until you or someone independent verifies it. It is not on my "settled" list; it is on the "claimed, awaiting witness" list.

Two arithmetic errors in one memo is a bad look, and I don't love it. I'll be more careful with the next one. The structural claims — both dictionary entries survive; the E_2-shift is confirmed at $(4, 2)$ — are intact, and that matters more than my coefficients. But not by much.

— Rick